

# Transcendental Algebraic Geometry

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**Abstract**

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# 1

## 1.1

Recall that the category of schemes and the category of analytic spaces are both full subcategories of locally ringed spaces.

### Definition 1.1.1 (The Analytification of a Scheme)

Let  $X$  be a scheme locally of finite type over  $\mathbb{C}$ . Define the analytification of  $X$  to be the unique analytic space  $X^{\text{an}}$  together with a morphism of locally ringed space  $\alpha : X^{\text{an}} \rightarrow X$  satisfying the following universal property: For any analytic space  $Y$  and any morphism of locally ringed spaces  $f : Y \rightarrow X$ , there exists a unique morphism of analytic spaces  $g : Y \rightarrow X^{\text{an}}$  such that the following diagram commutes:

$$\begin{array}{ccc} & Y & \\ \exists! g \swarrow & \downarrow f & \\ X^{\text{an}} & \xrightarrow{\alpha} & X \end{array}$$

Recall that the Yoneda embedding  $h : \mathbf{LRS} \rightarrow \mathbf{Set}^{\mathbf{LRS}}$  is the functor sending a locally ringed space  $X$  to the functor  $h_X : \mathbf{LRS} \rightarrow \mathbf{Set}$  defined by  $h_X(Y) = \text{Hom}_{\mathbf{LRS}}(Y, X)$ . Then the universal property of the analytification means that the functor  $h_X$  restricted to analytic spaces  $\mathbf{An}$  is representable.

One thinks of the morphism  $X^{\text{an}} \rightarrow X$  as doing two things. It is an enlargement of the standard topology of  $X^{\text{an}}$  to the Zariski topology of  $X$ , and it introduces generic points outside of  $X(\mathbb{C})$  (Indeed we will see that  $X^{\text{an}}$  has the same underlying set as  $X(\mathbb{C})$ ).

### Theorem 1.1.2

Let  $X$  be a scheme locally of finite type over  $\mathbb{C}$ . Then the analytification of  $X$  exists and is unique up to unique isomorphism.

### Definition 1.1.3 (The Analytification Functor)

Define the analytification functor

$$(-)^{\text{an}} : \mathbf{Sch}_{\mathbb{C}} \rightarrow \mathbf{An}$$

as follows.

- For each scheme  $X$  locally of finite type over  $\mathbb{C}$ , we have  $X^{\text{an}}$  is the analytification of  $X$ .
- For a morphism of schemes  $f : X \rightarrow Y$  over  $\mathbb{C}$ , the universal property of  $Y^{\text{an}}$  induces a morphism  $X \rightarrow Y^{\text{an}}$ . Precomposing with the canonical morphism  $X^{\text{an}} \rightarrow X$  gives the desired map  $X^{\text{an}} \rightarrow Y^{\text{an}}$ .

### Proposition 1.1.4

Let  $X$  be a scheme locally of finite type over  $\mathbb{C}$ . Then the underlying set of  $X^{\text{an}}$  is given by  $X(\mathbb{C})$ .

### Proposition 1.1.5

Let  $X$  be a scheme locally of finite type over  $\mathbb{C}$ . Then the following are true.

- If  $X$  is separated, then  $X^{\text{an}}$  is Hausdorff.
- If  $X$  is proper, then  $X^{\text{an}}$  is proper.

### Proposition 1.1.6

Let  $X$  be a variety over  $\mathbb{C}$ . Then the following are true.

- $X$  is complete if and only if  $X^{\text{an}}$  is compact.
- If  $X$  is smooth, then  $X^{\text{an}}$  is a complex manifold.
- If  $X$  is irreducible, then  $X^{\text{an}}$  is connected.

**Theorem 1.1.7 (Grothendieck)**

Let  $X$  be a smooth variety over  $\mathbb{C}$ . Then there is a natural isomorphism

$$H_{\mathrm{dR}}^n(X) \cong H_{\mathrm{Holo}}^n(X^{\mathrm{an}}) \cong H_{\mathrm{dR}}^n(X^{\mathrm{an}})$$

induced by the canonical map  $\alpha : X^{\mathrm{an}} \rightarrow X$ .

**Corollary 1.1.8**

Let  $X$  be a smooth variety over  $\mathbb{C}$ . Then the above isomorphism and de Rham's theorem induces a natural isomorphism

$$H_{\mathrm{dR}}^n(X) \cong H_{\mathrm{sing}}^n(X^{\mathrm{an}}; \mathbb{C})$$

**Theorem 1.1.9 (The Period Isomorphism)**

Let  $k$  be a field. Let  $X$  be a smooth variety over  $k$ . Then there is a natural isomorphism

$$H_{\mathrm{dR}}^*(X) \otimes_k \mathbb{C} \cong H_{\mathrm{sing}}^*(X^{\mathrm{an}}; \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$$

**1.2****Proposition 1.2.1**

Let  $X$  be a smooth projective curve over  $\mathbb{C}$ . Then  $X^{\mathrm{an}}$  is homeomorphic to  $\Sigma_{p_g(X)}$  the orientable surface of genus  $p_g(X)$ .